

Oleksii Odarchuk

Backend Developer (Go) · Creator of rombik and the Piton language



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Backend engineer (Go) with a product mindset and half a year of commercial experience. 1st at BEST Hackathon 2026, 3rd at Mate Hackathon, invited to DOU Day 2026. From hackathon MVPs to production Go services.

EXPERIENCE

Backend Developer (Go)

October 2025 — present

SkyService

- Design and build **architecturally complex Go microservices from scratch** for a production platform: interface-driven (hexagonal) architecture, inter-service communication, work under real production load.
- Designed an AI orchestrator — a web analog of claude-code**: a control-plane / execution-plane split — Claude runs through a tool-use loop while **all agent code executes inside isolated Docker sandboxes**. Core guarantee, **enforced by an automated test**: secrets (Anthropic key, git tokens) **never reach the sandbox**. MCP-over-SSE, credential-free git (GitHub App: JWT → installation-token), an **egress-allowlist proxy** against exfiltration, a container-lifecycle supervisor (idle-reaper, orphan-recovery).
- AI support-bot platform (RAG)**: semantic memory on PostgreSQL + pgvector (text / vector / hybrid search), a self-learning knowledge graph, an MCP server, schema-per-tenant isolation, token-based billing.

WINS & HACKATHONS

1st Place — Lead Backend & Sensor Fusion

April 18-19, 2026

BEST Hackathon 2026 — GPS-less Drone Trajectory

- Algorithm to **reconstruct a UAV's trajectory without GPS** from IMU/barometer logs. Sensor fusion in **Go + C**: Go for the pipeline & integration, C for the hot numerical kernels.
- 1st place among top teams** — judges singled out the architectural resilience: modular split telemetry / integration / simulation let us swap datasets without rewriting.

3rd Place — Backend Engineer

March 14, 2026

Mate Hackathon (50 participants, 10 teams)

- Designed and built a Go backend from scratch in **7 hours** for an AI-powered MarTech tool that automates competitor-ad analysis.
- Dual AI pipeline: GPT-4o for marketing trigger analysis + Nano Banano 2 (fal.ai) for creative generation; backend routed all AI calls to cloud APIs under hackathon load. Beat teams with 5+ years of experience.

Security Researcher → Job Offer

2026

College Blockchain System "Student Hryvnia"

- Discovered and exploited a critical vulnerability: the **backend never checked transaction amounts** (only the frontend) → arbitrary negative-value transfers. Reported responsibly — received a job offer.
- Reverse-engineered the official JS miner via Webpack bundles — then a faster Go replacement (SHMiner — in Projects).

DOU Day 2026 — personal invitation

Personally invited by the organisers to the conference with a free ticket — while still a student, ahead of professional work.

TECH STACK

Languages

GoPythonCTypeScript

Backend

GinFiberRESTWebSocketsSSEGoroutinesJWTOAuth 2.0

Data

PostgreSQLpgvectorRedisClickHouse

Frontend

SvelteWailsTailwind

AI

Anthropic ClaudeOpenAIMCPRAGfal.ai

DevOps

DockerKubernetesLinux (Arch)GitTinyGoTypst

PROJECTS

FLAGSHIP · COMMERCIAL micro-SaaS

website → API →

◆ rombik — Flowchart Generator

My first full-fledged commercial micro-SaaS · code in 10 languages → a DSTU 19.701-90 / ISO 5807 flowchart or structogram

Designed and run entirely by me. A **Go engine** — a compiler pipeline (parser → IR → layout → render) on **tree-sitter** for **10 languages** (Python, JavaScript, TypeScript, C, C++, C#, Java, Go, PHP, Pascal); flowcharts and Nassi-Shneiderman structograms, pixel-for-pixel to DSTU 19.701-90 / ISO 5807, byte-exact golden tests. Web editor (pan/zoom, lasso-splitting), 8 formats (SVG/PNG/PDF/Word·Visio·draw.io as native shapes/Typst/Excalidraw). **Billing runs on my own go-monobank-sdk** (HMAC webhooks, grants, receipts). Compiler craft carried over from my language **Piton**. Public **API**, **CLI**, **MCP server**, an AI skill and describe→flowchart via AI. Free watermarked PNG, a clean export needs an account.

Go · Gin tree-sitter API · CLI · MCP go-monobank-sdk PostgreSQL DSTU

OPEN SOURCE SDK

go-monobank-sdk — github →

Go SDK for every monobank API

Full coverage of **personal**, **provider** (with monoКЕП), **business** for legal entities, **acquiring** (subscriptions, monipay button, split-receivers, T2P) and **Installment Purchases**. ECDSA webhook verification with auto-rotating keys, drop-in http.Handler, typed money/currency/MCC, HMAC-SHA256 for installment callbacks, a fake monobank server for tests, and a separate OpenTelemetry sub-module.

Go ECDSA HMAC-SHA256 OpenTelemetry iter.Seq2

SDK CONSUMER

monokasa — Telegram github →

Ticket-Selling Bot

Sells tickets to a single show through a monobank jar and issues PDFs with QR codes scanned at the venue entrance. Live consumer of my own **go-monobank-sdk**: webhook auto-registers at startup, reservations get a 15-minute hold, and show-time reminders fire at-most-once even across restarts.

Go SQLite telebot PDF + QR go-monobank-sdk

1000× FASTER

SHMiner — Desktop github →

Crypto Miner

Cross-platform desktop miner built with **Wails (Go + Svelte/TypeScript)**. Rewrote SHA-256 hashing from JS to Go with a goroutine worker pool — achieved **1000× faster hashrate** than the official client on identical hardware. AES-encrypted wallet storage, real-time WebSocket dashboard, multi-wallet, zen mode.

Go Svelte TypeScript Wails WebSockets Goroutines

150+ USERS

AbitAssistant Bot

[github →](#)

Telegram bot helping Ukrainian students track university admission rankings. Reverse-engineered closed third-party APIs to fetch live competition data. **150+ active users in first year**, marketing campaign planned for 2026.

Python

Aiogram

PostgreSQL

BeautifulSoup

Pandas

Docker

LinkShortener Bot

[github →](#)

Telegram URL shortener with QR generation and geo analytics. PostgreSQL for storage, Redis for redirect caching, ClickHouse for real-time click analytics (country, city, platform). Docker, GeolIP2.

Go

PostgreSQL

Redis

ClickHouse

Docker

Telegram API

MY OWN LANGUAGE

Piton — My own programming language

[github →](#) [docs →](#)

Ukrainian transliterated language + interpreter, from scratch in **Go without yacc/bison**: recursive descent parser, tree-walking evaluator, Turing-complete. Flowcharts via a built-in **D2**-based engine. Interpreter compiled to **WebAssembly via TinyGo** (220 KB).

Go

TinyGo

WebAssembly

D2

Nix



EDUCATION & LANGUAGES

Network Technologies

IT College NULP, 2nd year
2024 — present

IT Class

Science Lyceum, Rivne
2022 — 2024

Ukrainian — native
English — B1